

BESIII X(2370) Glueball as Pure Vacuum Self-Binding - Fifth Tractor of Active Vacuum Proof of $Z_0=376.73\Omega$ AETERNVM VACUVM V6.8

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GitHub: <https://github.com/gustavosouzaconde40-ai/aeternvm-vacuum-rigidity-kappa>

Zenodo Primary: <https://zenodo.org/records/22949508> | Related: 22873164, 22651450, 22837427, 22821262, 22850558

Abstract

We present the isolated fifth tractor of the active vacuum proof chain centered on $Z_0 = \sqrt{\mu_0/\epsilon_0} = 376.730313\Omega$. The BESIII X(2370) state ($M=2370$ MeV, $J^{PC}=0^{--}$) discovered in 10 billion J/psi radiative decays and presented in Natal 2026, satisfies all four glueball criteria: mass ~ 2.4 GeV predicted by lattice QCD, quantum numbers 0^{--} , flavor-singlet blindness, and K^* anti- K^* decay cancellation. We interpret X(2370) as the first laboratory realization of pure vacuum self-binding: matter without valence quarks, where 100% of mass originates from self-coupled field tension ($E=mc^2$). In AETERNVM VACUVM framework, this is Z-Bits geometrically saturated (χ -field), self-attractive due to color charge, analogous to Z_0 as natural impedance unit from FAST-10P. Unlike photon, gluon carries its own charge [PIPA 03:46]. The 50-year mixture problem (IOTA/eta1440, $f_0(1500)$, $f_0(1710)$, χ_{2230}) is resolved by statistical depletion of 10B events, same logic as Conde Ruler $Z'=\text{gap}/\ln(p)$ immutable null model (mean 1.00041294). The flavor-singlet cancellation in K^* anti- K^* [PIPA 18:47] proves Z_0 -blind coupling: Z_0 has no flavor. This extends the four-tractor road under review at RAA (RAA-2026-0669, 26-Sep-2026, China version focused on LHAASO 3.73 PeV $\tau_{AV}=0.81$ vs $\tau\sim 4.5$ and FAST) to five tractors, preserving RAA manuscript distinct. Code MIT, TRL 3-4. China-VO PaperData / ScienceDB compliance retained.

Keywords

vacuum impedance Z_0 - glueball - BESIII X(2370) - active vacuum - QCD - dark energy - Conde Ruler

1. The Battle for Z_0

The central thesis of AETERNVM VACUVM is $Z_0 = \sqrt{\mu_0/\epsilon_0} \approx 376.73\Omega$ as active vacuum rigidity. FAST-10P measurement defines $S_{\text{inst}}=280 \Rightarrow k=2\pi Z_0/S_{\text{inst}}=8.45\Omega$. Conde Ruler $I=\log(1+\rho/\rho_0)$ defines $\Lambda_{\text{eff}}(Z_0, \nabla I)=Z_0 f(|\nabla I|)$ recovering LCDM. All tractors converge to Z_0 .

2. Four-tractor base (RAA-2026-0669 frozen)

(1) Conde Ruler, (2) FAST-10P Z_0 , (3) Six evidences: JWST CEERS high-z excess, IXPE PD=0.556, LHAASO Cygnus X-3 3.73 PeV transparency, LZ 2.6sigma, (4) V6.7.1 window I_F in $[\pi^2, 4\pi^2]$, $c_s^2=0.8$, I_2/I_1 0.22% residual. This version is frozen under RAA review.

3. Fifth tractor: BESIII X(2370)

3.1 Removal of matter leaves a particle [PIPA 00:00-02:14]: 99% of proton mass is binding energy. Removing quarks leaves self-bound gluons. 3.2 Self-charge [03:46-05:12]: gluons carry color, attract each other, form bound state without quarks [07:06]. Lattice QCD predicts 2.4 GeV [08:48]. 3.3 50 years false alarms [09:27-12:13] due to quantum mixing of q - $\bar{q}$ mesons with same quantum numbers. 3.4 Discovery [12:45-16:07]: BESIII e^+e^- collider as gluon factory, J/psi radiative decay, X(2370) 2011 signal at 2370 MeV, 2024 10B decays 0^{--} confirmed. 3.5 Final proof: flavor singlet [16:46-19:46]: glueball cosmopolitan, flavor-blind, K^* anti- K^* channel cancellation $\rightarrow$ not seen in 10B events, impossible for normal meson. 3.6 Natal 2026 [20:45]: four criteria met.

4. Mapping to AETERNVM VACUVM

Common matter = residual tension of Z-Bits χ . Dark matter = geometrically saturated Z-Bits. Dark energy = macroscopic proportion derived from Z0. Glueball = saturated Z-Bits in QCD sector: pure χ -field vortex, mass from tension. Conde Triangle $P(k|n)=(1/2^n)\text{binom}(n,k)\rightarrow\text{Gauss}$ predicts low-entropy 0^{+} state count. VIEC-Mk.IV-C-FEMP-V1 $t_n=t_{n-1}+t_{n-2}$ (Dumitrescu Nature 2022) 50Ω line Al6061 slots $3\tau, 8\tau, 13\tau$, 3 barriers: Geometric, Algebraic Crouzeix-Jin $\|p(A)\|\leq 2$, Topological, is hardware analog of X(2370) memory.

5. Folder structure for GitHub aeternvm-vacuum-rigidity-kappa V6.8

/tractor-1-conde-ruler/ (conde-governante DOI 22651450 immutable 1M prime gaps)

/tractor-2-FAST-10P-Z0-376.73/ (S_inst=280, Z0 unit)

/tractor-3-JWST-CEERS-high-z/ (active vacuum large scale)

/tractor-4-LHAASO-3.73PeV-transparency/ (tau_AV=0.81)

/tractor-5-BESIII-X2370-glueball-pure-self-binding/ (this PDF + analysis + PIPA video ref <https://www.youtube.com/watch?v=obqUOnfKi-Y>)

/proof-Z0-376.73-common-denominator/ (convergence table JWST-LHAASO-BESIII-LZ-IXPE)

/releases/V6.8/

This structure keeps RAA China version separate and makes Z0 proof chain explicit.

6. Zenodo strategy

Primary upload: zenodo.org/records/22949508 NEW VERSION V6.8. Title: five-tractor... Related identifiers: IsNewVersionOf 22873164, IsSupplementTo 22651450, 22837427, 22821262, 22850558, References RAA-2026-0669. Other repos: AETERNVMVACUVM (22873164) stays legacy monolith archive; conde-governante (22651450) stays null model only; VACUO-ATIVO-6-PROVAS (22837427) links as evidence compilation; V631-FAST-10P (22821262) stays Z0 instrument; ENGINE (22850558) stays lab. All point to kappa repo as current development head.

Data Availability

China-VO PaperData and ScienceDB per RAA. Zenodo DOIs: 10.5281/zenodo.22949508 (V6.8 primary), 22873164, 22651450, 22837427, 22821262, 22850558. Code: <https://github.com/gustavosouzaconde40-ai/aeternvm-vacuum-rigidity-kappa> - MIT. BESIII data: BESIII Collaboration 2011-2024 10B J/psi events. Video: PIPA <https://www.youtube.com/watch?v=obqUOnfKi-Y>